

Combined Stress Analysis with Strain Rosettes & Mohr's Circle

Measuring normal and shear stress at six points on an aluminum beam under combined bending and shear

Author	Luis Ramirez (with team)	Performed	April 17, 2025
Course	MECH 237A Strength of Materials Lab, NJIT	Instructor	Prof. Milano · TA: Oscar Poudel
Team	Kacper Czarnecki, Ashlee Hercules, Luis Ramirez, Jackson Robillard	My role	Individual discussion and conclusion; data analysis with team
Equipment	Aluminum beam load frame, delta and rectangular strain rosettes, strain indicator, Excel		

PROJECT SUMMARY

This experiment used strain rosettes and Mohr's circle to evaluate combined stresses in a simply supported aluminum beam under coupled bending and shear. Six rosettes recorded strain at different locations along the beam, and the team converted those strains into normal and shear stresses using plane-stress relations, then built experimental and theoretical Mohr's circles for each point. Rosettes 1–3, located away from the applied load, agreed best with theory, with normal stresses within 2–11% of predictions. Rosettes 4–6, near the concentrated load, showed larger deviations due to local stress concentrations and placement sensitivity, and rosette 4 appeared to be faulty. The results confirm stress-transformation theory under ideal conditions while showing where simple beam models lose accuracy.

1. Objectives & Theory

The goals were to measure stresses at several points on the beam, compare them with predictions from bending and shear theory, and use Mohr's circle to visualize and evaluate stress transformation.

Two rosette types were used: **delta rosettes** with gauges at 0°, 60°, and 120°, and **rectangular rosettes** with gauges at 0° and ±45°. After the strains ϵ_x , ϵ_y , and γ_{xy} were resolved from the gauge readings, stresses were found using Hooke's law for plane stress:

$$\sigma_x = E/(1 - \nu^2) \cdot (\epsilon_x + \nu\epsilon_y), \quad \sigma_y = E/(1 - \nu^2) \cdot (\epsilon_y + \nu\epsilon_x), \quad \tau_{xy} = G\gamma_{xy}$$

Theoretical stresses came from the flexure formula $\sigma = My/I$ and the shear formula $\tau = VQ/Ib$. Each state of stress was then plotted as a Mohr's circle, centered at $(\sigma_x + \sigma_y)/2$ with radius $\sqrt{[(\sigma_x - \sigma_y)/2]^2 + \tau_{xy}^2}$, from which the principal stresses $\sigma_{\max}$ and $\sigma_{\min}$ are read directly.

2. Results

Table 1. Theoretical vs. experimental stresses at each rosette (σ_y is zero in theory for rosettes 1–3)

Rosette	Quantity	Theoretical (psi)	Experimental (psi)	Exp / Theo
1	σ_x	-1,875	-1,667	0.89
	τ_{xy}	469	909	1.94
	σ_{max}	111	375	3.39
	σ_{min}	-1,986	-2,072	1.04
2	σ_x	0	45	—
	τ_{xy}	625	1,000	1.60
	σ_{max}	625	1,030	1.65
	σ_{min}	-625	-971	1.55
3	σ_x	1,875	1,835	0.98
	τ_{xy}	469	900	1.92
	σ_{max}	1,986	2,208	1.11
	σ_{min}	-111	-334	3.02
4	σ_x	-3,750	-1,949	0.52
	σ_y	-1,415	-1,175	0.83
	τ_{xy}	469	94	0.20
	σ_{max}	-1,324	-1,164	0.88
	σ_{min}	-3,841	-1,960	0.51
5	σ_x	0	306	—
	σ_y	-354	-1,041	2.94
	τ_{xy}	625	251	0.40
	σ_{max}	473	351	0.74
	σ_{min}	-826	-1,087	1.31
6	σ_x	3,750	3,568	0.95
	σ_y	-157	631	—
	τ_{xy}	469	-16	—
	σ_{max}	3,805	3,568	0.94
	σ_{min}	-213	631	—

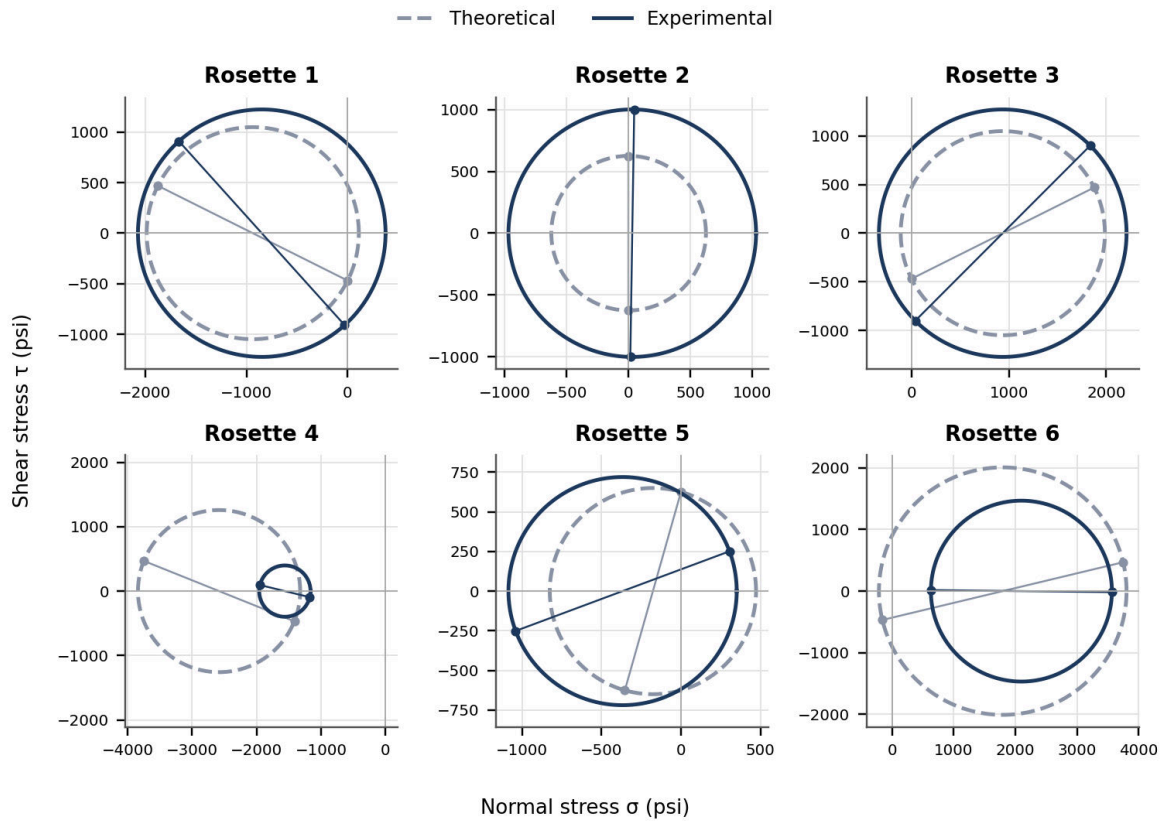


Figure 1. Theoretical (dashed) and experimental (solid) Mohr's circles for all six rosettes.

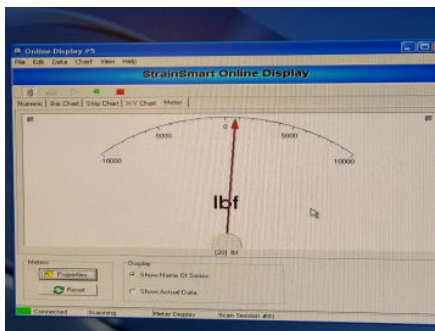


Figure 2. Strain indicator, unloaded

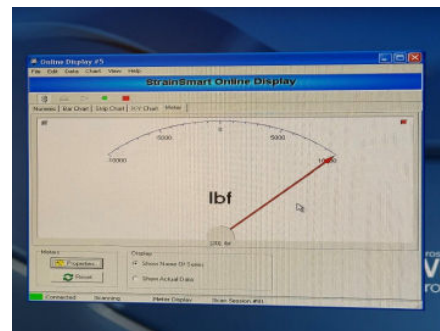


Figure 3. Strain indicator, loaded

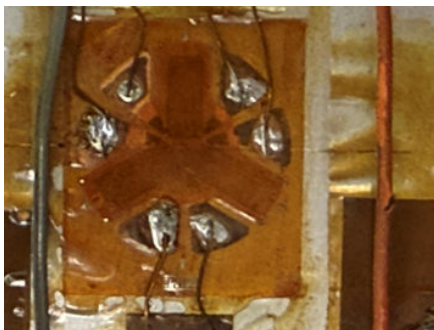


Figure 4. Delta (120°) strain rosette

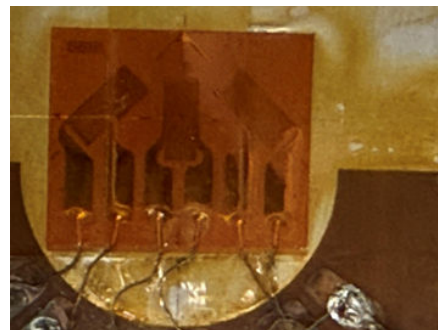


Figure 5. Rectangular (45°) strain rosette

3. Discussion

Rosettes 1–3 (away from the load). These rosettes sat in regions dominated by bending, and their experimental circles lined up well with theory in position. The normal stress σ_x was within 11% of prediction at rosette 1 and within 2% at rosette 3, and the minimum principal stress at rosette 1 was within 4.3%. The main difference was circle size: measured shear stress ran about 1.6–1.9 times the theoretical value, which enlarged the experimental circles. Even so, these three points gave the closest agreement with theory.

Rosettes 4–6 (near the load). These rosettes were close to the concentrated load, where the simple beam formulas are least accurate. Stresses there are affected by local stress concentrations and by the vertical bearing stress under the load, and small differences between a rosette's actual and intended position cause large changes in the reading. Rosette 6 still matched σ_x and $\sigma_{\max}$ within about 6%, but rosette 5 showed a 31% error in $\sigma_{\min}$, and rosette 4 showed no meaningful relationship to theory, with the shear stress at only 20% of its predicted value. Because the other rosettes performed much better, rosette 4 most likely had a defect or bonding problem.

Nonzero σ_y . Theory assumes $\sigma_y = 0$ away from the load, but small nonzero values were measured at rosettes 1–3. Real conditions are never ideal: slight misalignment, minor unevenness in the beam, and small bending effects that cannot be fully removed all contribute. These values were small and had little effect on the overall stress state.

Sources of error. The most likely sources are rosette placement and alignment, imperfections in the beam, local stress concentrations near the load, and human error during setup and reading.

4. Conclusion

The experiment met its objectives: stresses were calculated at every rosette using strain transformation, and experimental and theoretical Mohr's circles were compared. Under conditions close to ideal, away from the applied load, the flexure and plane-stress theories agreed well with measurement, confirming stress transformation and Mohr's circle as reliable tools. Near the concentrated load, larger deviations exposed the limits of simple beam models. Together, these results show why experimental validation matters: theory provides a strong foundation, but real materials under complex loading need to be measured to understand how they actually respond.

References

- Ciesla, M. (2017). *Strength of materials laboratory manual*. New Jersey Institute of Technology.
- Beer, F. P., Johnston, E. R., Jr., DeWolf, J. T., & Mazurek, D. F. (2020). *Mechanics of materials* (8th ed.). McGraw Hill.